

MASTER'S THESIS

Framing, Time Pressure, and Environmental Attitude in Sustainable Consumption Decisions

Dawei Yang · Marketing Management · Data Marketing and Analytics

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Abstract

This study examines whether framing effects in sustainable consumption remain observable in realistic online decision environments characterized by limited attention, heterogeneous engagement, and noisy behavioral data. Prospect theory predicts that loss framing should increase environmentally sustainable choices, yet empirical evidence outside laboratory settings is often unstable. The central question is whether framing operates as a general mechanism or whether its influence depends on context and individual processing conditions.

To address this issue, the study combines framing with cognitive and environmental constraints in a scalable online design. A 2×2 between-subject experiment manipulates gain versus loss framing and instruction-based time pressure. Participants complete repeated binary consumption decisions, generating decision-level panel data that allow analysis of both average effects and within-individual variability. Environmental attitude is measured before the task as a directional anchor. Reaction time is used as a behavioral indicator of processing intensity, and decision stability is constructed from repeated choices as an exploratory boundary condition.

The results suggest that the average framing signal is directionally consistent but statistically unstable under clustered inference. Loss framing tends to increase sustainable choice, yet the magnitude varies across specifications and individuals. Time pressure, implemented through instruction rather than strict deadlines, does not generate a uniform behavioral constraint. Instead, perceived urgency appears heterogeneous and engagement-dependent. This weak and imprecise amplification effect indicates that time-based interventions in digital environments rely on credibility and attention rather than formal design alone.

Environmental attitude emerges as the most stable predictor of baseline behavioral orientation. Its moderating role is directionally consistent but imprecise. Reaction-time analysis shows that loss framing is associated with longer decision time, suggesting increased processing demands even when choice effects remain modest. At the same time, the operational conflict index is associated with shorter response times, indicating that it does not function as a clean proxy for processing intensity in realistic online contexts.

Exploratory analysis further indicates that instability in average framing partly reflects heterogeneity in decision stability. Framing sensitivity appears concentrated among less stable individuals, whereas more stable decision-makers show weaker responses to contextual variation. This boundary-condition pattern provides a structural explanation for

why framing effects appear unstable in realistic and heterogeneous environments.

1. Introduction

1.1 Background and the Real-World Problem

Despite technological progress, environmental pressure remains persistent. Climate change, biodiversity loss, and pollution continue to affect public health and daily life. These issues have been widely discussed, and significant advances have been made in policy, technology, and public awareness. As a result, environmental protection is no longer framed solely as a technical or institutional challenge. It increasingly appears in everyday decision contexts.

This study does not focus on large-scale policy or technological solutions. Instead, it addresses a narrower question with practical relevance: why do consumers often ignore environmental considerations in real consumption settings? More specifically, why does sustainable consumption remain difficult to promote in everyday decisions?

1.2 Traditional Economic Perspective: Stable Preferences and Constraints

From a traditional economic perspective, sustainable consumption is primarily understood as the outcome of rational choice under preference heterogeneity and external constraints. Standard rational choice theory assumes that individuals possess stable, well-defined, and internally consistent preferences (Samuelson, 1947). Whether environmental outcomes are valued depends on the utility weight individuals assign to them. Some consumers hold strong pro-environmental preferences and are willing to pay higher costs for sustainability, whereas others prioritize price, convenience, or functionality.

If preferences are stable, choices should not systematically change across different descriptions or contexts as long as the feasible choice set remains unchanged. In this framework, higher prices for environmentally friendly products reduce demand, but price enters the utility maximization problem only as a trade-off. Consumers with sufficiently strong environmental preferences are still expected to select sustainable products.

Similarly, incomplete information is treated as a cost rather than a structural barrier. In standard economic models, consumers are assumed to collect information until the marginal benefits of search equal its marginal costs. Complex labels or certification systems therefore increase search costs and influence choice probabilities, but they do not alter the underlying decision mechanism. Time is also treated as a scarce resource in economic

decision-making. Individuals allocate time optimally, and even under time pressure the rational structure remains intact. Time constraints are interpreted as higher decision costs rather than a source of systematic bias. Overall, traditional economic theory explains limited sustainable consumption mainly through insufficient preference strength or changing constraints.

1.3 Tension with Reality and the Behavioral Turn

As income levels and access to information have improved, several constraints emphasized in traditional models—such as price barriers and information costs—have partly weakened. Under rational choice theory, this should lead to a gradual increase in sustainable behavior. In practice, however, progress has been slow, and the intention–behavior gap remains substantial.

The key assumption behind the traditional framework is preference stability rather than price or information alone. Environmental preferences are treated as consistent and exogenous. If this holds, choices should not vary across alternative descriptions of the same outcome. Framing theory challenges this directly. Tversky and Kahneman (1981) show that presentation can alter decisions even when objective information is unchanged. Preferences are therefore not always revealed but partly constructed.

Real-world decisions are embedded in language, context, and symbols. Risk communication, product labeling, and policy framing shape how individuals perceive outcomes. Environmental issues are typically long-term, uncertain, and loss-oriented. Loss framing may therefore increase sensitivity to future risks and shift attention toward long-term consequences. Two patterns emerge early and guide the interpretation of subsequent results.

1.4 External Validity of Framing Effects

Most framing research is conducted in relatively controlled environments. In contrast, real consumption settings are complex and dynamic. Real consumption environments are cognitively unstable. For instance, in supermarkets or online platforms, it is difficult to compare all alternatives systematically. As a result, individuals rely on heuristics or habitual responses. Time pressure is not always an intentional choice. It is frequently shaped by the decision environment.

In addition, real decision making increasingly takes place in digital and online contexts. Information updates rapidly, and choice processes are often unstable. Behavioral data show

substantial variability. This raises an important question: does framing remain effective in realistic decision environments? Is its impact contingent on available cognitive resources? Do individuals with different environmental attitudes respond differently to framing information?

1.5 Research Positioning: Cognitive Constraints and Realistic Decision Environments

This study integrates framing with cognitive constraints and focuses on realistic decision contexts. The central goal is to examine whether framing effects depend on cognitive resources. Specifically, time pressure is introduced as a key constraint in order to test how decision behavior changes under different levels of available cognitive capacity. Rather than assuming a universal framing effect, this study treats framing as a conditional mechanism shaped by cognitive resources and individual differences.

1.6 Time Pressure as a Cognitive Constraint

Many framing studies implicitly assume that individuals have sufficient time and attention to process information. When cognitive resources are limited, however, the decision process may change. Real decision making is not based on unlimited rationality. Instead, it is constrained by available resources, among which time is the most direct and universal factor.

1.7 Heterogeneity and Environmental Attitude

To capture individual differences, this study introduces environmental attitude as a pre-existing preference anchor. Environmental attitude does not directly determine behavior. Instead, it reflects value orientation and preference strength. This variable allows the identification of heterogeneity in responses to framing and time pressure, providing a way to examine how stable value structures interact with cognitive constraints.

1.8 Decision Processes and Reaction Time as Behavioral Indicators

As discussed above, framing may interact with time pressure and influence not only choices but also the decision process. A basic decision path can be described as follows: exposure to information, reaction time, cognitive conflict, and final choice. Even when information is understood quickly, a measurable response time is still required. For this reason, the present study focuses on both choices and reaction times in order to model decision processes more

precisely.

Dual-process theories distinguish between fast, heuristic processing (System 1) and slow, analytical processing (System 2). Different frames may activate different processing routes. However, internal cognitive mechanisms cannot be directly observed. In behavioral research, reaction time is therefore widely used as an indicator of processing intensity. Longer response times typically suggest higher cognitive conflict and deeper evaluation.

1.9 Decision Stability, Behavioral Variability, and Realistic Choice

Individual decisions are not always stable across time and contexts. Behavioral variability is common. In traditional models, such variability is often treated as noise. Behavioral research, however, interprets variability as meaningful, reflecting cognitive conflict, unstable preferences, or fluctuations in attention. Framing effects may therefore not be uniform across individuals. Instead, they may be concentrated among low-stability individuals.

This study introduces decision stability as a behavioral dimension derived from repeated trial-level behavior but conceptualised as a participant-level construct. Stability is treated as an observed behavioral feature rather than an exogenous causal factor. It may nevertheless shape decision pathways.

1.10 Online Decision Environments and Noisy Behavioral Data

Behavioral variability becomes more pronounced in real environments. Both online and offline consumption involve dynamic information flows. Individuals face fragmented and rapidly changing inputs, which create heterogeneous time pressure. To investigate this mechanism, experimental environments must simulate such conditions.

For this reason, the present study emphasizes structured data pipelines, reproducibility, and decision-level analysis. Instead of relying solely on clean aggregate data, this approach captures the dynamics of realistic choice behavior and improves ecological validity.

1.11 Research Contributions

This study contributes to the literature by re-examining framing effects from the perspective of cognitive constraints and external validity. While framing has been widely documented in controlled laboratory settings, much less is known about its robustness in realistic and dynamic decision environments. Rather than assuming stability, this study focuses on whether framing signals remain directionally observable when attention is fragmented and

engagement is heterogeneous. By introducing time pressure as a structural feature of decision environments, the analysis shifts attention from average effects toward conditional relevance.

A second contribution lies in integrating time pressure into sustainable consumption research as a practical and observable constraint. Unlike abstract manipulations of cognitive load, time pressure is embedded in everyday consumption contexts such as online shopping and promotional deadlines. The present design prioritizes scalability and feasibility over strict experimental control. This makes it possible to examine how behavioral interventions function in environments that more closely resemble digital markets, where urgency is perceived rather than mechanically enforced.

The study also incorporates environmental attitude as a directional anchor. Instead of treating preference strength and behavioral mechanisms as competing explanations, the framework allows both to operate simultaneously. Stable values define baseline orientation, while contextual signals shape short-run behavioral variation. This makes it possible to identify when framing is more or less likely to translate into observable behavior, rather than assuming uniform effects.

Another contribution is methodological. The joint analysis of choice and reaction time extends traditional outcome-based approaches and provides evidence on processing dynamics. Reaction time is treated as an observable indicator of processing intensity rather than a precise measure of internal states. This process-oriented perspective allows the study to detect behavioral signals even when average choice effects are weak or unstable.

The study emphasizes structured data management and decision-level analysis in online behavioral research. Instead of over-cleaning heterogeneous data, the analytical pipeline preserves variability and treats it as potentially informative.

1.12 Project Objectives

This project aims to examine whether framing effects in sustainable consumption remain directionally observable but empirically unstable in realistic online decision environments characterized by cognitive constraints, heterogeneous attention, and noisy behavioral data. Rather than testing framing under ideal laboratory conditions, the goal is to evaluate its robustness and practical relevance in scalable and digitally mediated settings.

The first objective is to assess whether loss framing increases the probability of sustainable choice in a repeated and cognitively demanding decision environment. This provides a

benchmark test of whether framing remains detectable when behavioral variability, bounded attention, and heterogeneous engagement are present.

The second objective is to examine whether perceived time pressure, implemented as instructional urgency rather than strict enforcement, influences decision behavior and alters responsiveness to contextual signals in realistic online environments. This reflects real consumption environments in which decision speed is often constrained by task structure rather than formal deadlines.

The third objective is to integrate preference heterogeneity into the analysis. Environmental attitude is introduced as a directional anchor that captures baseline orientation toward sustainability. The project evaluates whether stable value structures influence behavioral consistency and baseline choice orientation, and whether they condition responsiveness to contextual variation.

The fourth objective is methodological. The project develops a decision-level behavioral dataset and a transparent, reproducible data preparation pipeline that preserves variability rather than over-cleaning observations. This approach emphasises structured data construction, minimal exclusion, and analytical transparency, reflecting the role of data analytics in transforming noisy online behavioral data into interpretable evidence.

Finally, the project explores whether decision stability functions as a boundary condition of framing sensitivity. Using entropy-based measures derived from repeated choices, the analysis examines whether framing effects are concentrated among individuals with less stable decision patterns. This exploratory objective does not aim to replace established behavioral theories but to identify boundary conditions in realistic decision environments.

Together, these objectives position the study as both a behavioral investigation and a data-driven project aimed at improving the practical relevance of behavioral insights in sustainable consumption and digital decision contexts.

2. Literature Review

2.1 Prospect Theory

Traditional economic models of decision making under risk are grounded in expected utility theory, which assumes stable and internally consistent preferences across equivalent representations. Within this framework, individuals evaluate outcomes in terms of final wealth states and should remain invariant to alternative descriptions of objectively identical options. As a result, behavioral differences are interpreted as reflecting variations in preferences or constraints rather than differences in cognitive representation (Samuelson, 1947).

However, a large body of empirical evidence has challenged this view. Prospect theory, proposed by Kahneman and Tversky (1979), offers a descriptive account of decision making under risk. Instead of assuming evaluation based on final states, the theory suggests that individuals encode outcomes relative to a reference point. This implies that the same objective outcome may be perceived differently depending on how it is framed.

A central mechanism of prospect theory is reference dependence. Individuals classify outcomes as gains or losses relative to a reference level rather than evaluating absolute consequences. In addition, loss aversion plays a fundamental role. Losses tend to have a stronger psychological impact than equivalent gains, which generates systematic asymmetries in risk preferences. Individuals often display risk-averse behavior in the gain domain but become more risk-seeking when facing potential losses.

These insights provide a micro-level explanation for deviations from rational invariance. If outcomes are evaluated relative to a reference point, then equivalent information presented in gain or loss terms may produce different preferences. In the context of sustainable consumption, environmental decisions often involve long-term uncertainty and potential future losses. Therefore, loss-based representations of environmental outcomes may be particularly effective in motivating behavioral change.

2.2 Framing Effect

Building on prospect theory, the framing effect refers to systematic shifts in preferences caused by differences in the presentation of equivalent information. Tversky and Kahneman (1981) demonstrate that decision outcomes can vary even when objective consequences remain unchanged. This finding suggests that preferences are constructed during the

decision process rather than fully pre-determined.

Subsequent research has clarified that framing is not a single phenomenon. Levin, Schneider, and Gaeth (1998) distinguish among risky-choice framing, attribute framing, and goal framing. These categories differ in their cognitive mechanisms and empirical implications. Risky-choice framing typically involves probabilistic outcomes, attribute framing focuses on descriptive valence, and goal framing concerns the consequences of action or inaction. This typology helps explain why empirical results across studies are often heterogeneous.

Framing has also been widely applied in environmental and risk communication. Spence and Pidgeon (2010) find that loss-framed environmental messages can increase perceived climate risk and motivate engagement. However, empirical findings regarding behavioral outcomes remain mixed. In some contexts, framing significantly affects decisions, while in others the effect is weak or unstable. These inconsistencies suggest that framing effectiveness depends on contextual and psychological factors rather than being universally robust.

Taken together, existing research indicates that framing effects are theoretically well established but empirically conditional. An important unresolved question concerns the boundary conditions under which framing influences real-world behavior. Identifying these conditions is particularly relevant in complex consumption environments where cognitive resources may be constrained.

2.3 Dual-Process Theory and Cognitive Resources

Dual-process theories provide an important framework for understanding how cognitive constraints shape decision making. These theories distinguish between intuitive and automatic processing (System 1) and slower, deliberative processing (System 2) (Evans, 2008). System 1 relies on heuristics and affective responses, whereas System 2 involves analytical evaluation and cognitive control. The balance between these two systems depends on the availability of cognitive resources, including time and attention.

Situational constraints such as cognitive load, distraction, and time pressure can shift individuals toward more intuitive processing. When cognitive resources are limited, individuals are less likely to engage in systematic evaluation and more likely to rely on simplified decision strategies. This perspective implies that behavioral biases, including framing effects, may vary depending on processing conditions.

Time pressure represents a direct and ecologically relevant constraint in consumer decision making. Prior research shows that time constraints influence decision strategies and may lead consumers to simplify choices or defer decisions (Maule and Svenson, 1993; Dhar and Nowlis, 1999). These findings suggest that time pressure alters the decision process rather than merely increasing decision costs. Therefore, framing effects may not be stable across contexts but may depend on the level of available cognitive resources.

Accordingly, time pressure provides a practical way to examine how cognitive constraints shape the impact of framing in realistic environments.

2.4 Sustainable Consumption and Contextual Constraints

Sustainable consumption has become an important domain for studying decision making under complexity and uncertainty. A large body of research documents a persistent gap between pro-environmental attitudes and actual purchasing behavior. Carrington et al. (2014) show that ethical intentions often fail to translate into real consumption behavior, especially in complex and realistic settings. This gap suggests that contextual and situational factors play an important role beyond stable preferences.

Traditional explanations emphasize structural barriers such as price and product availability. However, recent research highlights the role of decision environments and psychological cues. Consumers often face multiple goals, limited attention, and complex product information, which encourages reliance on heuristics rather than systematic comparison.

Behavioral interventions, such as nudges, aim to influence sustainable consumption by modifying the decision context. For example, Cornelissen et al. (2008) demonstrate that highlighting common environmentally friendly behaviors can increase sustainable choices. Similarly, communication strategies emphasizing environmental risks can influence perception and motivation (Spence and Pidgeon, 2010).

Despite these advances, evidence on the behavioral impact of framing in sustainable consumption remains mixed. This inconsistency indicates that situational and cognitive constraints may moderate the effectiveness of framing. Therefore, sustainable consumption provides a suitable context to examine how cognitive constraints interact with framing in real-world decision environments.

2.5 Environmental Attitude and Preference Heterogeneity

Individual heterogeneity represents another key factor influencing decision outcomes.

Environmental attitude reflects the strength and stability of pro-environmental preferences and has been widely linked to sustainable behavior (Stern, 2000). However, preferences do not operate in isolation. They interact with situational and cognitive factors.

Individuals with strong environmental values are more likely to maintain consistent choices and resist contextual influences. In contrast, individuals with weaker attitudes may rely more on external cues and heuristics. This implies that framing effects may be moderated by underlying preference structures.

Empirical evidence supports this view. Peters and Levin (2008) show that individual differences in cognitive abilities influence susceptibility to framing. These findings suggest that framing effects are likely to vary across individuals depending on underlying preference strength and cognitive engagement. In other words, framing effects are heterogeneous rather than uniform.

In this study, environmental attitude is introduced as a pre-existing preference anchor to capture heterogeneity in decision stability. Rather than directly predicting behavior, it is expected to moderate the relationship between framing, cognitive constraints, and choice. This approach integrates traditional preference-based explanations with behavioral mechanisms and allows a more comprehensive understanding of sustainable consumption.

2.6 Data Management, Online Experiments, and Decision-Level Analysis

Goodman et al. (2013) show that online samples can provide valid and cost-effective data when appropriate design and screening procedures are applied. Similarly, attention checks and structured monitoring improve data reliability in online environments (Hauser and Schwarz, 2016).

An important challenge concerns the preparation and organization of behavioral data. Online environments generate heterogeneous and noisy observations. Rather than treating variability solely as error, recent work suggests that such variation may reflect meaningful differences in attention, engagement, and decision strategies (Hertwig and Ortmann, 2001). Therefore, structured data preparation and transparent analytical workflows are necessary to preserve behavioral information while ensuring interpretability.

In this study, a decision-level panel structure is adopted to integrate behavioral choices with reaction-time indicators. The focus is not on algorithm development but on transforming raw experimental output into decision-relevant evidence. This approach reflects the role of data analytics in managerial contexts, where analysts coordinate data preparation, quality control,

and interpretation to support decision making.

3. Conceptual Framework

3.1 Cognitive Representation and Framing

Framing does not change objective information. The alternatives remain identical in terms of price and environmental impact. What changes is how individuals encode outcomes. In this sense, framing operates at the level of cognitive representation rather than preference itself. When environmental information is presented as a gain, individuals focus on positive outcomes or improvements. When it is presented as a loss, attention shifts toward risk and potential deterioration.

Environmental problems are naturally characterized by long-term uncertainty and irreversible consequences. These features are more easily activated under loss-based descriptions. Loss framing increases the salience of future risks and strengthens sensitivity to potential negative outcomes. As a result, the same objective trade-off may be perceived as more urgent or consequential.

This perspective emphasizes representation rather than bias. Framing is treated as a mechanism that shapes how decision-relevant information enters the cognitive system. From this view, the directional prediction in this study follows directly: loss framing should increase the probability of sustainable choice in realistic decision environments.

3.2 Decision Process and Cognitive Activation

Decision making unfolds over time rather than occurring as a single step. Individuals encounter external signals, integrate them with internal orientation, and then generate responses. During this process, framing may either align with existing preferences or introduce tension.

This tension does not only influence outcomes. It also affects the structure of processing. When signals are aligned, responses may be rapid and automatic. When misaligned, additional evaluation or hesitation may occur. As a result, both final choice and response time carry information about decision dynamics.

Reaction time is therefore treated as an observable indicator of processing intensity. Even when information is understood quickly, measurable time is required. Longer responses may reflect deeper evaluation or temporary hesitation, whereas shorter responses may signal rapid alignment or disengagement. This approach extends the analysis beyond outcomes

and provides a basis for examining how conflict and processing interact in realistic environments.

3.3 Processing Mode and Dual-Process Mechanism

Differences in decision outcomes can be understood through variation in processing modes. Rather than a strict classification, dual-process theory is interpreted here as a description of alternative cognitive pathways. Some decisions rely on intuitive and rapid processing, while others involve more deliberate evaluation.

Framing effects are more likely to emerge when intuitive processing dominates. In such situations, individuals respond to salient representations rather than checking informational equivalence. Analytical processing, in contrast, increases attention to trade-offs and reduces sensitivity to presentation.

From this perspective, framing is not a stable effect. Its influence depends on which processing pathway is activated. The same individual may switch between modes across situations. This explains why framing results are often heterogeneous and context-dependent. The central implication is that framing operates through shifts in processing regimes rather than through a constant behavioral bias.

3.4 Cognitive Constraint and Time Pressure

Real-world decision environments are characterized by limited cognitive resources. Individuals rarely evaluate all available information in a stable and systematic manner. Instead, attention is fragmented and decision time is constrained. Among different forms of resource limitation, time pressure is one of the most direct and observable.

Time pressure influences how information is processed. Under constrained conditions, individuals are less likely to engage in extended deliberation and more likely to rely on simplified representations. This may increase reliance on salient contextual cues such as framing. Conversely, when time is available, individuals may engage in deeper evaluation and reduce sensitivity to contextual variation.

In this study, time pressure is treated as a structural feature of decision environments rather than a control variable. It captures the speed and urgency embedded in many real consumption contexts, especially online platforms. This perspective strengthens external validity and provides the theoretical basis for examining whether framing effects depend on available cognitive resources.

3.5 Preference Stability and Environmental Attitude

Individual differences represent an important source of heterogeneity in behavioral responses. Environmental attitude is introduced as a directional anchor reflecting the strength and stability of pro-environmental preferences. Rather than determining behavior directly, it shapes the baseline orientation of decision making.

Individuals with strong environmental attitudes are more likely to display consistent sustainable choices. Their internal value structure provides a stable reference, reducing reliance on contextual cues. In contrast, individuals with weaker attitudes may show greater variability and depend more on external signals. These mechanisms are not mutually exclusive and may coexist within the same decision context.

This framework integrates preference-based and behavioral explanations. Stable preferences define long-run orientation, while contextual mechanisms shape short-run variation. Environmental attitude therefore functions as a stabilizing force that may buffer framing sensitivity without eliminating contextual influence.

3.6 Decision Stability and Behavioral Variability

Observed behavior is often variable across time and situations. Rather than treating this variability as noise, this study interprets it as a meaningful behavioral feature. Variability may reflect cognitive conflict, fluctuations in attention, or unstable construction of preferences.

Decision stability is introduced as a behavioral dimension capturing consistency across repeated choices. Individuals differ in how consistently they translate preferences into action. Some display stable patterns, while others show greater sensitivity to context.

From this perspective, framing effects may be concentrated among individuals with lower stability. Stable decision makers rely more on internal representations and show weaker responses to contextual variation. Stability therefore functions as a potential gating mechanism that may determine whether external signals influence behavior. This approach moves beyond average treatment effects and provides a structural explanation for heterogeneous and unstable framing results.

3.7 Integrated Model

The conceptual framework integrates representation, processing, and preference. Framing shapes cognitive representation and activates risk-related concerns. This activation may generate alignment or conflict with internal preferences.

Time pressure constrains processing and influences which cognitive pathway dominates. Under constrained conditions, intuitive processing becomes more prominent, increasing reliance on contextual cues. Environmental attitude provides a directional anchor that stabilizes choice orientation. Decision stability captures the consistency with which internal preferences are translated into action.

Together, these mechanisms determine both behavioral outcomes and processing dynamics. The final structure links contextual representation, cognitive constraints, and preference stability to observable choice and reaction time.

4. Hypotheses

Based on the theoretical framework and research questions outlined above, the following hypotheses are proposed.

4.1 H1. Loss framing increases sustainable choice.

In sustainable consumption contexts, loss-framed messages are expected to increase the likelihood of choosing environmentally friendly products compared to gain-framed messages. This hypothesis is grounded in prospect theory, which predicts that individuals evaluate outcomes relative to a reference point and exhibit stronger sensitivity to potential losses than equivalent gains.

In environmental decision-making, loss framing highlights the potential negative consequences of unsustainable behavior, making future risks more salient. This increased salience is expected to shift attention toward long-term outcomes, thereby increasing the probability of selecting sustainable options. The goal is not to test whether framing works under ideal laboratory conditions, but whether a directional framing signal remains observable in a realistic and noisy online environment characterized by heterogeneous attention and incomplete engagement.

$$\text{Pr}(\text{Choice} = 1 \mid \text{Frame} = 1) > \text{Pr}(\text{Choice} = 1 \mid \text{Frame} = 0)$$

Formally, Choice = 1 indicates selecting the sustainable option, and Frame = 1 indicates the loss condition. This hypothesis serves as the baseline benchmark for evaluating framing under realistic decision constraints.

4.2 H2. Decision conflict systematically affects processing time.

Decision conflict is expected to arise when externally presented framing cues are misaligned with individuals' internal environmental attitudes. In such situations, individuals must integrate competing representations, increasing the likelihood that decision processes deviate from automatic or heuristic responding.

Importantly, in this design conflict is a participant-level construct mechanically derived from pre-task attitude and between-subject framing assignment. It therefore captures cross-individual differences in attitude–frame misalignment rather than within-individual trial-to-trial fluctuations. Reaction time is therefore interpreted as a behavioral proxy for momentary processing demands rather than fixed cognitive speed.

Formally, conflict reflects the deviation between framing direction and environmental attitude:

Conflict = $|\text{attitude}_z - \text{frame_dir}|$, where $\text{frame_dir} = +1$ (Loss) and -1 (Gain)

Higher conflict is expected to alter deliberation dynamics. In constrained or realistic online environments, this influence may operate through multiple processing pathways. In some cases, conflict may increase deliberation and slow responses; in other cases, it may trigger rapid heuristic alignment or disengagement. The key prediction is therefore that conflict **systematically modulates processing time**, rather than producing a strictly monotonic increase.

This formulation allows the hypothesis to be tested under realistic online conditions where cognitive strategies, attention, and engagement vary substantially across individuals. Because conflict is constant within individuals in this design, the analysis focuses on how baseline preference–frame misalignment influences repeated decision processing rather than trial-level fluctuations.

4.3 H3. Time pressure amplifies framing effects.

Time pressure is expected to increase reliance on contextual cues by reducing available cognitive resources for systematic processing. When deliberation is constrained, individuals are more likely to depend on salient representations rather than fully integrating all available information.

Framing provides a salient contextual cue that can guide rapid decision-making. Therefore, under time pressure, loss framing is expected to exert a stronger influence on sustainable choice compared to low-pressure conditions. This prediction is consistent with resource-based models of decision-making, which suggest that heuristic processing becomes more dominant when cognitive capacity is limited.

In realistic online environments, however, time pressure is often perceived rather than mechanically enforced. The present design therefore focuses on perceived urgency induced by instructions rather than strict time limits. Under such conditions, amplification effects are expected to depend on the **intensity and credibility of perceived constraints**, as well as heterogeneity in attention and task engagement. As a result, the magnitude of this effect may vary across individuals and contexts.

4.4 H4. Environmental attitude buffers framing sensitivity.

Environmental attitude is expected to reduce susceptibility to framing by providing a stable internal reference. Individuals with strong pro-environmental attitudes are more likely to rely on their existing value orientation rather than external contextual cues.

Consequently, framing effects are predicted to be weaker among individuals with stronger environmental attitudes.

Environmental attitude is conceptualized as capturing the **directional component of preferences**, distinguishing individuals with stronger pro-environmental orientation from those with weaker or more neutral positions. This stabilizing role may operate through both directional strength and consistency of preferences, allowing internal representations to dominate over contextual framing in decision-making.

4.5 Exploratory extension: Stability as a boundary condition of framing

Beyond the original hypotheses, preliminary observations suggested that the average framing effect may not be uniform across individuals. Instead, framing sensitivity may depend on the stability of underlying preferences.

Decision stability is measured using entropy derived from repeated behavioral choices. Unlike self-reported attitudes, this measure reflects revealed consistency across multiple decisions and may capture latent decisional thresholds and the reliability of internal preference structures. In repeated-choice environments, individuals with unstable or weakly consolidated preferences may rely more on contextual cues, while those with stable decision patterns may exhibit reduced sensitivity to framing.

This approach is particularly relevant in online experiments, where repeated behavioral data provide richer information than single-shot preference elicitation and allow direct observation of decision consistency.

These analyses are exploratory and intended to generate future hypotheses rather than test pre-registered predictions. The goal is to examine whether framing effects are concentrated among individuals with less stable decision patterns, while remaining weak or absent among those with more stable preferences.

Importantly, this analysis does not replace the original theoretical framework but seeks to identify boundary conditions under which framing is more or less effective in realistic online

decision environments.

4.6 Summary of theoretical structure

The hypotheses together define a process-based framework linking contextual representation, cognitive processing, and behavioral outcomes. Framing influences cognitive representation, which may generate conflict when external cues and internal preferences diverge. Time pressure constrains deliberation and increases reliance on salient contextual information. Environmental attitude provides a directional anchor for preferences, while decision stability captures the consistency and decisional threshold underlying repeated choices. This framework allows the study to test both average effects and conditional heterogeneity in a realistic online environment characterized by bounded attention and heterogeneous engagement.

5. Methodology

5.1 Experimental design

The study adopts a 2×2 between-subject experimental design manipulating framing (Gain vs. Loss) and time pressure (Low vs. High). The goal is not to isolate framing under ideal laboratory conditions but to examine whether framing remains observable in a realistic and cognitively constrained online decision environment.

Participants were randomly assigned to one of the four conditions. Randomization provides identification of the causal effects of framing, time pressure, and their interaction on sustainable choice. Each participant completed 24 repeated binary decisions between a sustainable (eco-friendly) and a conventional product. Product categories and trial order were randomized to reduce learning and ordering effects.

This repeated-decision structure generates a decision-level panel dataset, which allows the study to examine both average effects and within-individual behavioral variation. The use of multiple trials also provides richer behavioral information than single-shot elicitation and enables the construction of stability measures as an exploratory extension.

5.2 Participants and recruitment

Participants were recruited through online channels and were required to be at least 18 years old and capable of understanding the task instructions. The experiment was conducted in English and simplified Chinese to increase accessibility and ecological validity in a multilingual online setting.

The initial sample included all respondents who provided informed consent and entered the experimental task. Following pre-specified quality criteria, participants were excluded only if they did not complete the experiment or failed to produce valid behavioral choices. This minimal exclusion strategy was adopted to preserve heterogeneity and avoid over-cleaning, which could distort behavioral patterns.

The final dataset contains both participant-level and decision-level observations.

5.3 Procedure

The experimental procedure followed a structured online workflow designed to balance realism, scalability, and internal validity.

Participants first completed a consent form and basic demographic questions. Environmental attitude items were then administered before exposure to the experimental manipulation. This ordering ensured that the measurement of preferences was not contaminated by framing or time pressure.

Participants were subsequently randomly assigned to one of the four experimental conditions. In the high time-pressure condition, participants were instructed to make decisions as quickly as possible due to limited time. In the low-pressure condition, no such instruction was provided. This instruction-based manipulation was chosen due to platform constraints and scalability considerations. While weaker than automated countdown methods, it allows implementation in large-scale online environments.

The effectiveness of this manipulation was evaluated behaviorally by comparing reaction times across pressure conditions.

Following the manipulation, participants completed 24 binary consumption decisions. Each trial presented a product choice framed either in gain or loss terms, depending on the assigned condition. The order of trials was randomized.

An attention and recall check was embedded within the trial sequence and randomized across positions. This design allows quality control while preserving the natural decision flow. However, given the multilingual context, attention failure was treated as a robustness lever rather than a strict exclusion criterion. This approach recognizes that misunderstanding may reflect language processing rather than disengagement. In this context, attention failure is interpreted as part of bounded attention and heuristic responding in realistic online decision environments rather than pure disengagement, and therefore serves as a robustness filter rather than a primary exclusion criterion.

5.4 Measures

5.4.1 Choice

The primary dependent variable is sustainable choice. For each trial, participants selected either an eco-friendly or a conventional product. The variable is coded as:

- **Eco choice = 1 if the sustainable option is selected, and 0 otherwise.**

This decision-level structure allows analysis of both average treatment effects and within-participant behavioral dynamics.

Framing was treated as a categorical variable in all models, with Gain as the baseline category and Loss as the treatment condition. This specification follows the experimental design and allows direct interpretation of framing effects relative to a reference condition.

5.4.2 Reaction time

Reaction time (RT) was recorded using the Qualtrics timing function, capturing the duration each decision page remained active. RT was log-transformed to reduce skewness and to stabilise variance.

RT is interpreted as a behavioral proxy for processing demand rather than a direct measure of latent psychological states. The objective is to capture processing dynamics while allowing both within-individual and cross-individual variation. Only valid decision trials were included in the RT analysis. Extreme values were preserved in the baseline dataset to retain information about behavioral variability and cognitive effort. As a result, observed reaction-time variation reflects both situational processing and stable individual differences rather than purely momentary cognitive states.

5.4.3 Environmental attitude

Environmental attitude was measured using multiple Likert-scale items capturing concern about environmental issues. The items were combined into a composite index after checking internal consistency. The composite score was standardized (attitude_z) and treated as a participant-level variable.

Environmental attitude is interpreted as the directional component of preferences. It reflects whether individuals are more or less inclined toward sustainability but does not assume direct alignment between stated attitudes and behavior.

Environmental attitude was standardized (z-score) to improve numerical stability in interaction models and ensure comparability of coefficients across specifications. All subsequent analyses therefore use the standardized measure (attitude_z).

5.5 Conflict index

To operationalize the cognitive mechanism underlying framing, a participant-level conflict index (misalignment proxy) was constructed. The index captures the degree of misalignment between externally presented framing cues and internal environmental orientation.

framing direction was coded as a binary signal, with Loss framing mapped to +1 and Gain

framing mapped to -1. This coding is arbitrary and does not affect interpretation, as the conflict index is defined as an absolute deviation.

Conflict was defined as the absolute difference between standardized environmental attitude and framing direction:

- **conflict = |attitude_z - frame_dir|**

Standardisation ensures that internal preference orientation and external framing signals are measured on comparable scales, improving interpretability of the conflict index.

This measure is constructed mechanically using pre-task attitude and randomized framing, rather than inferred from behavioral outcomes such as reaction time. It therefore reflects exogenous variation in decision tension induced by the interaction between internal preferences and external cues.

The central prediction is that higher conflict increases processing demand and thus reaction time. However, in realistic online and multilingual decision environments, this operational proxy may not produce a strictly monotonic increase in response time. Conflict is therefore treated as a behavioral construct subject to measurement noise and heterogeneous processing strategies rather than a definitive mediator.

5.6 Decision stability

Beyond the original hypotheses, the study examines whether framing sensitivity depends on the stability of individual preferences.

Decision stability was computed using entropy based on repeated choices. This participant-level measure captures cross-individual differences in the consistency of observed behavior. Unlike self-reported measures, it reflects revealed patterns of decision-making.

Because stability is derived from observed behavior, it is interpreted descriptively rather than as an exogenous trait. Stability measures were standardized and evaluated across alternative specifications to ensure robustness. The analysis therefore focuses on whether framing effects vary across stability levels rather than treating stability as a causal determinant.

Stability is not included in the baseline causal models but is instead used in heterogeneity and boundary-condition analyses. This design avoids post-treatment bias while allowing the

study to examine when framing effects emerge in realistic decision environments.

5.7 Data construction and reproducibility

The data pipeline followed a structured and reproducible workflow.

Raw Qualtrics data were first exported and cleaned by removing metadata rows. The dataset was then converted from wide to long format, with each row representing a single decision. This transformation enabled panel-style analysis and improved alignment with experimental behavioral research.

Participant-level environmental attitude and stability measures were merged into the decision-level dataset. The final dataset includes key variables such as participant ID, trial, framing, time pressure, product category, choice, reaction time, language, environmental attitude, and stability.

Data and outputs were stored in structured directories, including raw data, cleaned datasets, model results, and figures. This pipeline facilitates transparency and replicability.

5.8 Estimation strategy

5.8.1 Choice models

The main analyses were conducted at the decision level using logistic regression models with participant-clustered standard errors to account for within-individual correlation arising from repeated choices.

This approach allows flexible modelling of heterogeneous behavioral responses while avoiding strong distributional assumptions required by hierarchical models. Standard errors were clustered at the participant level to ensure robust inference under the panel structure.

Primary models test framing, time pressure, and their interaction. Language was included as a control variable in extended specifications to account for systematic differences in information processing across linguistic environments. Preliminary descriptive analysis indicated systematic differences in decision speed and baseline behavior across languages, motivating this control.

Environmental attitude was introduced as a moderator to examine whether stable preference direction buffers framing sensitivity. Although language is treated primarily as a control

variable in the main models, exploratory analyses also examine its potential moderating role in shaping framing sensitivity and processing behavior.

5.8.2 Reaction time models

Reaction time analyses use linear models on log-transformed RT. The main specification examines the relationship between conflict and processing time while controlling for time pressure and framing.

Language was also included because preliminary descriptive analysis indicated systematic differences in decision speed across linguistic environments. These models test whether cognitive tension induced by misalignment between internal preferences and external cues is reflected in observable decision time rather than self-reported difficulty.

5.8.3 Robustness and validation

Robustness analyses were pre-specified and focused on three dimensions.

First, attention and recall failure were treated as a robustness filter rather than a strict exclusion rule. This approach reflects the view that inattentive or heuristic responding is a feature of real-world decision environments rather than purely noise.

Second, extreme reaction times were trimmed in sensitivity analyses to evaluate whether main findings depend on distributional outliers.

Third, environmental attitude was split into high and low groups to verify that buffering patterns remain directionally consistent.

Additional checks examined whether the stability boundary condition is sensitive to alternative behavioral consistency measures and subsample definitions.

5.8.4 Model specification

The main decision model is estimated at the trial level using logistic regression:

- $\text{Pr}(\text{eco_choice_it} = 1) = \text{logistic}(\beta_0 + \beta_1 \text{Loss_it} + \beta_2 \text{Pressure_it} + \beta_3 (\text{Loss_it} \times \text{Pressure_it}) + \beta_4 \text{attitude_z_i} + \beta_5 (\text{Loss_it} \times \text{attitude_z_i}) + \epsilon_{it})$.

where i indexes individuals and t indexes decision trials.

Loss and Pressure are between-subject treatment variables that remain constant within

individuals. The variable `attitude_z` is measured at the participant level and merged into the trial-level dataset.

Extended specifications additionally include language as a control variable:

- **β_6 Language_i**

Because the dataset has a repeated decision structure, observations are not independent within individuals. Therefore, all models are estimated with standard errors clustered at the participant level.

The reaction time model is specified as:

$$\log(\text{RT}_{it}) = \gamma_0 + \gamma_1 \text{Conflict}_i + \gamma_2 \text{Pressure}_{it} + \gamma_3 \text{Loss}_{it} + \gamma_4 \text{Language}_i + u_{it}$$

The error term allows for within-individual correlation.

Exploratory extensions further examine whether framing sensitivity varies across levels of behavioral stability constructed from repeated decisions. These analyses are interpreted as boundary-condition evidence rather than primary causal tests.

5.9 Experimental Implementation

This study was implemented using the Qualtrics online survey platform. The purpose of this section is to document the practical construction of the experiment, focusing on how stimuli, randomization, and behavioral measurement were operationalized under real platform constraints while maintaining the identification logic of behavioral economic experiments.

5.9.1 Platform and Survey Flow

The experiment was structured through the Qualtrics Survey Flow, which governed the sequencing of modules and the assignment of experimental conditions. Embedded Data fields were used to store treatment indicators and key variables throughout the survey. Participants first completed informed consent, demographic questions, and the environmental attitude scale before entering the randomization stage.

Participants were assigned in equal proportions to four conditions defined by Frame (Gain vs. Loss) and Time Pressure (High vs. Low). These variables were written into Embedded Data and subsequently used to control question visibility through display logic. This implementation ensured a strict between-subject design while reducing the possibility of

exposure to multiple framing conditions. Each participant was exposed to only one framing condition throughout the experiment.

5.9.2 Time Pressure Manipulation

Because the survey was conducted under account and platform restrictions, no automatic countdown or forced time limit was implemented. Instead, time pressure was induced through instruction. Participants assigned to the high-pressure condition were shown a short instruction page before entering the decision task, informing them that they should make their decisions quickly. No such instruction was presented in the low-pressure condition.

This approach is commonly used in online behavioral research when technical enforcement of response time is not feasible. It is intended to induce perceived time pressure rather than mechanically restrict decision time. The effectiveness of this manipulation was evaluated behaviorally by comparing reaction times across pressure conditions. In this setting, reaction-time differences are interpreted as evidence about the *strength and heterogeneity* of the manipulation (i.e., whether and for whom the instruction translated into faster decisions), rather than as a strict manipulation “pass/fail” criterion.

Given this instruction-based design, the manipulation is expected to generate heterogeneous perceived urgency rather than a uniform reduction in processing time.

5.9.3 Multi-Trial Decision Task

The core behavioral task consisted of repeated binary choices between an eco-friendly and a conventional product. Each participant completed 24 trials. The task was implemented using the Loop & Merge function, allowing the experimental structure to generate a trial-level decision sequence.

The Loop & Merge table included the following fields:

- product category
- percentage change in emissions
- percentage change in price
- attention-check indicator

These values were dynamically inserted into the decision text in each trial. Although the structure of the task remained constant across participants, the order of trials was randomized. This reduces order effects and discourages rule-based responding.

The product scenarios focused on everyday consumption items in order to increase familiarity.

5.9.4 Framing Implementation

The gain and loss conditions differed only in linguistic framing and reference point. All objective information—price and emission trade-offs—remained identical. The design followed the principle of informational equivalence.

Multiple versions of the decision question were implemented within the same page and controlled through display logic. In each trial, only the version consistent with the assigned condition was visible. This ensured that framing differences were confined to language rather than information content. No moral or emotionally charged expressions were introduced, and the structure of the decision problem remained constant.

This approach is commonly used in behavioral research.

5.9.5 Attention Check

An attention check was embedded within the trial sequence and randomized across positions. Display logic ensured that it appeared only in the designated trial. Participants were required to select a specific option in order to proceed.

Embedding the check within the decision sequence allows quality control while maintaining the natural flow of the task.

5.9.6 Reaction Time Measurement

Reaction time was measured using the Qualtrics Timing Question. This tool records page-level response duration and was placed on the decision page within each loop. The measure therefore captures the combined time required for reading and responding. It should be interpreted as a proxy for processing time rather than precise click-level latency.

This approach is widely used in online experiments where custom scripting is unavailable. The focus here is not absolute timing precision but relative differences across experimental conditions.

5.9.7 Quality Control During Experiment

Several safeguards were implemented during data collection. Participants who did not provide informed consent could not proceed. All decision questions required a response,

preventing missing behavioral data. The attention check was mandatory.

The survey structure was tested internally before deployment to ensure stability in randomization and data recording. No additional exclusion rules were applied during data collection; screening was conducted during the analysis stage.

5.10 Data Preparation and Management

This section describes the structured workflow used to convert the Qualtrics export into an analysis-ready behavioral dataset. The focus is not on technical system development but on data organization, quality control, and analytical readiness. In data analytics and managerial environments, the key task is to coordinate data preparation processes and ensure that behavioral information can support reliable decision making.

5.10.1 Raw data and platform characteristics

The data were exported from Qualtrics in CSV format. As is typical in online survey platforms, the export includes both variable identifiers and descriptive labels. The dataset therefore required initial cleaning to remove non-observational rows and to ensure consistency across variables. The original raw file was preserved unchanged, and all subsequent work was conducted on structured working versions to ensure traceability.

Because the experiment used a repeated-choice design, the raw data were initially organized in a participant-level format. This structure is suitable for survey reporting but not for behavioral analysis. Therefore, the first step was to reorganize the data into a decision-level format, where each row represents a single choice. This structure allows the analysis of decision dynamics and heterogeneity.

5.10.2 Structured workflow and analytical governance

To ensure transparency and consistency, a structured workflow was adopted. The scope of analysis was defined in advance, and a working dataset was frozen before formal modelling. This approach reduced analytical flexibility and improved the credibility of the results.

The workflow focused on three objectives:

1. Transform raw experimental output into a structured behavioral dataset.
2. Maintain behavioral variability rather than over-cleaning the data.

3. Ensure that all steps are transparent and reproducible.

This process reflects the role of data analysts in coordinating and supervising data preparation rather than performing technical implementation.

5.10.3 Variable construction and interpretation

Key variables were constructed to support behavioral interpretation. The primary outcome variable is sustainable choice. Environmental attitude was measured before the task and combined into a composite index to capture directional preference.

Decision stability was constructed from repeated behavioral observations to reflect revealed consistency. This measure is interpreted as a behavioral indicator rather than a fixed individual trait. Reaction time was retained as an indicator of processing intensity. Extreme values were preserved to capture variation in decision strategies.

5.10.4 Quality monitoring and robustness

A conservative quality strategy was adopted. Only incomplete cases were excluded. Other indicators, such as attention failures and extreme response times, were retained as diagnostic information. This approach recognizes that real decision environments involve bounded attention and heterogeneous engagement.

This strategy preserves meaningful variation and improves external validity. It also reflects managerial analytics practices, where the goal is to understand behavioral patterns rather than to eliminate complexity.

5.10.5 Analytical readiness

The final dataset supports decision-level analysis and allows the examination of both average effects and behavioral heterogeneity. This structured preparation ensures that the data can be used to generate interpretable insights about consumer decision processes under realistic constraints.

Overall, this workflow demonstrates how online behavioral data can be organized and governed in a way that supports evidence-based decision making.

6. Results

6.1 Sample, structure, and basic patterns

The analysis is conducted at the decision level, with repeated trials nested within individuals. After minimal quality control, the choice models include 2,424 observations from 101 participants. Reaction-time models use 2,197 valid observations. The reduction reflects invalid or missing timing records in some trials. All inference relies on participant-clustered standard errors.

Two patterns emerge early and guide the interpretation of subsequent results. First, the framing coefficient is consistently positive but unstable once clustered inference is applied. Second, environmental attitude strongly predicts behavior, whereas the instruction-based time pressure manipulation does not produce a clear behavioral signature.

First, the average framing signal is directionally consistent but not stable: across specifications, the Loss coefficient is typically positive, but precision is limited under clustered SE. Second, there is strong evidence that pre-task environmental attitude is behaviorally relevant (it predicts choice robustly), while time pressure—as implemented via instruction—does not reliably shift reaction time, which matters for how strongly H3 can be supported in this online setting.

Table 1. Sample and experimental design overview

Category	Variable	N	Percentage
Participants	Total	101	100%
Observations	Total	2424	–
Mean trials	Average	24	
Framing	Gain	1320	54.50%
	Loss	1104	45.50%
Time Pressure	No	1248	51.50%
	Yes	1176	48.50%
Language	Chinese	1920	79.20%
	English	504	20.80%

Notes: Observations refer to decision-level data. Each participant completed 24 randomized trials. Percentages are calculated based on the total number of observations. The sample sizes reported in the figures correspond to the number of valid decisions in each condition.

Table 1 reports the sample structure and condition counts

6.2 H1 — Average framing effect on sustainable choice (Step 6A / 6B)

Note: For transparency, the analytical workflow is organised into sequential steps, with formal statistical testing beginning at Step 6 after data cleaning and exploratory diagnostics.

Step 6A (primary): $\text{eco_choice} \sim \text{frame}$

In the primary framing test with clustered SE, the estimated Loss effect is positive but not statistically significant:

- Loss vs Gain: OR = 1.56, 95% CI [0.82, 2.97], $p = 0.177$.

This result is not best interpreted as “no framing.” Instead, it indicates that in this realistic online setting the average framing effect is weakly positive but imprecise once within-subject dependence is taken seriously. The direction aligns with H1, but the data do not support a clean, universal average effect at conventional thresholds.

Step 6B (control for language): $\text{eco_choice} \sim \text{frame} + \text{language}$

Adding language as a control does not change the framing pattern materially (Loss remains positive, still imprecise), but language itself shows a clear baseline difference:

- Loss vs Gain: OR = 1.59, 95% CI [0.83, 3.03], $p = 0.160$.
- ZH-S vs EN: OR = 0.52, 95% CI [0.28, 0.97], $p = 0.041$.

So, at the level of average choice, the main shift when adding language is not a “rescued” framing effect; it is a baseline difference in eco-choice levels (ZH-S lower than EN), while the framing coefficient remains directionally positive but unstable. Importantly, language is treated here as a processing/measurement channel in a multilingual online task (comprehension and response dynamics), not as a cultural-value construct requiring substantive cultural interpretation. We do not interpret these differences as cultural value differences; they are reported as differences in processing dynamics within a multilingual online environment.

Interpretation constraint (important for later sections): after Step 6, the responsible claim is not “framing works” or “framing fails,” but: the average framing signal is present in direction but unstable in magnitude under clustered inference, which motivates looking at

mechanisms and boundary conditions rather than treating H1 as a single decisive pass/fail test.

Table 2. Logistic regression results on sustainable choice

Variables	Model (1)	Model (2)	Model (3)	Model (4)
Loss framing	1.56 (0.82 – 2.97)	1.59 (0.83 – 3.03)	1.89 (0.97 – 3.67)	1.89 (0.97 – 3.67)
Time pressure			1.57 (0.72 – 3.45)	1.49 (0.67 – 3.31)
Chinese language		0.52** (0.28 – 0.97)		0.54** (0.30 – 0.98)
Loss × Pressure			0.67 (0.24 – 1.86)	0.7 (0.25 – 1.96)
Observations	2424	2424	2424	2424

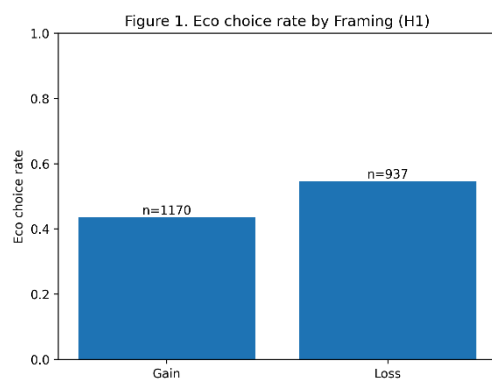


Figure 1. Sustainable choice under gain and loss framing

Notes: Odds ratios are reported. Clustered standard errors at the participant level. Confidence intervals are shown in parentheses. Model (2) and Model (4) additionally control for language. Interaction terms capture the moderating role of time pressure. **p < 0.05. (Report Step 6A and 6B in Table 2; Figure 1 can show predicted probabilities by frame, optionally with language as a control/overlay.)

The corresponding clustered logistic regression estimates are reported in Table 2, and Figure 1 provides the descriptive eco-choice rate by Framing condition.

6.3 H3 — Time pressure as a moderator of framing

Time pressure was implemented via instruction rather than automated countdown, prioritizing scalability and platform feasibility. Given this design choice, the key empirical question is whether the manipulation produces behaviorally meaningful constraints and whether it interacts with framing in choice. In realistic online decision environments, “time

pressure” is often perceived rather than mechanically enforced, so any behavioral signature may be modest and heterogeneous.

Descriptive patterns suggest that the instruction-based pressure manipulation may have generated heterogeneous perceived urgency across participants. However, formal behavioral validation based on reaction-time data is reported in Section 6.5. Any differences appear limited and unstable, which suggests that the instruction likely induced perceived urgency for some participants but did not translate into a consistent reduction in processing time at the aggregate level.

Step 7B (control for language): $\text{eco_choice} \sim \text{frame} * \text{time_pressure} + \text{language}$

Controlling for language, the interaction between framing and time pressure is not statistically distinguishable from zero in the clustered models. Instead, it suggests that perceived urgency is heterogeneous in online decision environments. Unlike laboratory countdown manipulations, instruction-based pressure may generate variation in engagement rather than a uniform constraint. As a result, amplification effects may be attenuated and unstable at the aggregate level. This interpretation is consistent with the modest reaction-time signature reported below and supports the view that the behavioral impact of time pressure depends on credibility, attention, and task engagement rather than the mere presence of urgency cues.

This is consistent with the RT models: pressure does not reliably reduce RT, which weakens the claim that the manipulation created a strong processing constraint in practice. (Report Step 7 results in Table 2 alongside H1, with the emphasis on interaction non-support and the practical limitation of manipulation intensity. Figure 2 can remain the Frame × Pressure plot, but it should be described as weak/unstable rather than definitive.) Behavioral signatures of perceived urgency are shown in Appendix Figure A5.

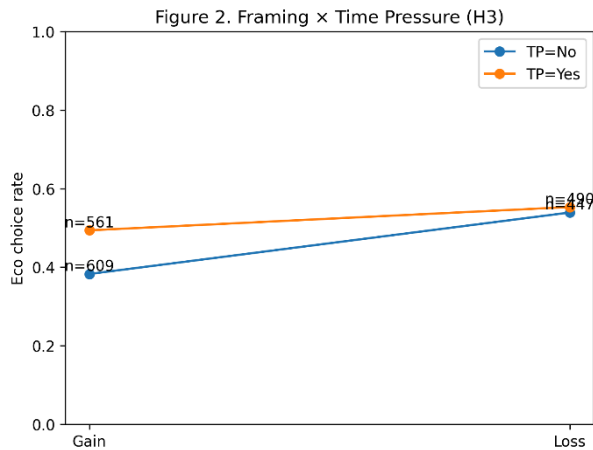


Figure 2. Sustainable choice across framing and time pressure conditions

Figure 2 visualises the descriptive pattern across the four cells; inferential results are reported in Table 2.

Notes: The figure reports descriptive eco-choice rates across framing and time-pressure conditions. Observations are at the decision level. Each participant completed 24 trials. Inference is based on logistic models with participant-clustered standard errors, reported in Table 2.

6.4 H4 — Environmental attitude as a buffer (Step 8)

Step 8 tests whether environmental attitude stabilizes choice and reduces reliance on external framing.

Main model: $\text{eco_choice} \sim \text{frame} * \text{time_pressure} + \text{language} + \text{attitude_z} + \text{frame}:\text{attitude_z}$

Two results stand out:

Attitude predicts choice strongly and robustly.

attitude_z: OR = 3.51, 95% CI approx. [2.14, 5.78], $p < 0.001$.

This is one of the most stable effects in the entire results set.

The buffering interaction is in the predicted direction but not supported with precision.

frame × attitude_z: OR = 0.74, 95% CI [0.37, 1.50], $p = 0.406$ (negative coefficient; imprecise).

So the clean conclusion for H4 is: attitude behaves as a strong directional anchor for sustainable choice, but evidence that it *specifically dampens* framing sensitivity is limited in this sample once clustered SE is used and other condition terms are present.

This matters for the overall story: attitude is not merely “noise.” It is a large and stable behavioral predictor, and it also provides a natural dimension along which instability in framing could arise (some participants are strongly anchored; others are not).

Table 3. Environmental attitude and sustainable choice

Variables	Model (5)
Loss framing	1.7 (0.63 – 4.64)
Time pressure	1.27 (0.54 – 3.03)
Chinese language	0.91 (0.43 – 1.94)
Loss × Pressure	1.13 (0.29 – 4.44)
Environmental attitude	3.51 (2.14 – 5.78)
Loss × Attitude	0.74 (0.37 – 1.50)
Observations	2424

Notes: Odds ratios are reported. Clustered standard errors at the participant level. Confidence intervals are shown in parentheses. Environmental attitude is standardized. ***p < 0.001.

(Report Step 8 in Table 3; any attitude-by-frame marginal plot should be described cautiously: directionally consistent buffering but wide uncertainty.)

6.5 H2 — Conflict and processing time (Step 9) + Pressure → RT (Step 10)

The RT analyses are intended as behavioral evidence about processing demand (not a literal readout of internal conflict). RT is log-transformed, and inference is clustered by participant.

Step 9: $\log RT \sim \text{conflict} + \text{time_pressure} + \text{frame} + \text{language}$

Two patterns appear:

- Frame (Loss) is associated with longer RT.
- coefficient = 0.274, $p = 0.005$

This is one of the clearer RT signals: Loss-framed trials take longer on average, consistent with framing increasing processing demands at least in some contexts.

- Conflict (as operationalized) is associated with *shorter* RT.
- coefficient = -0.183, $p = 0.001$.

Crucially, this does not force a collapse of the broader framing–processing link; it indicates that the *chosen operationalization* of “conflict” is not behaving as a clean processing-cost proxy in this implementation. For the remainder of the paper, conflict is therefore treated as an imperfect behavioral construct rather than a definitive mediator. Stability (entropy) is interpreted as a boundary-condition pattern that does not rely on any single conflict metric: stability may reflect a broader processing threshold rather than sensitivity to one operationalization of conflict.

Step 10: $\log RT \sim \text{time_pressure} + \text{language}$ (reduced specification focusing on the behavioral signature of the pressure manipulation).

Time pressure, implemented through instruction rather than deadlines, does not produce a statistically detectable reduction in reaction time. The estimated coefficient is small and imprecise (coef = 0.036, 95% CI [-0.175, 0.246], $p = 0.738$). This pattern indicates that perceived urgency was heterogeneous and did not translate into a uniform cognitive constraint at the aggregate level.

This result is informative rather than purely negative. In realistic online decision environments, time pressure is often perceived rather than mechanically enforced. The present findings therefore suggest that the effectiveness of time-based interventions depends not only on design intention but also on behavioral engagement and credibility. Under such conditions, amplification effects on framing may be limited and unstable.

Table 4. Reaction time models (log RT)

Variable	Model1(Step9)	Model2(Step 10)
Intercept	1.1375*** (0.1329)	1.0648*** (0.1256)
Time pressure (Yes)	- 0.0047 (0.0999)	0.0358 (0.1074)
Loss frame	0.2743** (0.0986)	
Language (ZH- S)	- 0.1208 (0.1138)	- 0.1649 (0.1174)
Conflict	- 0.1830** (0.0559)	
Observations	2197	2197
Clusters (participants)	101	101

Notes: The dependent variable is log-transformed reaction time. Clustered standard errors at the participant level are reported in parentheses.

*** $p < 0.001$, ** $p < 0.01$

6.6 Extension — Decision stability (entropy) as a boundary condition (Step 11)

The extension tests whether instability in average framing can be understood as heterogeneity in framing susceptibility across individuals with different decision stability. This is reported explicitly as exploratory boundary-condition evidence, not as a replacement of H1–H4. This extension is consistent with the earlier process framework in which baseline preference–context alignment and behavioral stability jointly determine when framing translates into observable behavior.

Additional descriptive evidence suggests that decision stability is also associated with systematic differences in processing time. As shown in Appendix Figures A6 and A7, more stable participants tend to exhibit longer average reaction times, consistent with the interpretation that stability may reflect higher decisional thresholds and sustained engagement rather than purely faster responding. These patterns are exploratory and should be interpreted cautiously given the sample size and distribution of stability in the current dataset.

Step 11 (continuous): $\text{eco_choice} \sim \text{frame} * \text{entropy_z} + \text{time_pressure} + \text{language} + \text{attitude_z}$

The model shows:

- **attitude_z remains a strong positive predictor (consistent with Step 8).**
- **The frame main effect is positive and near conventional significance, OR = 1.84, $p = 0.091$.**
- **The frame $\times$ entropy interaction is negative, OR ≈ 0.66 , $p = 0.218$.**

The direction of the interaction is consistent with a gating intuition—framing sensitivity decreases as stability increases—but the sample does not support a high-precision estimate under clustered SE. Visual evidence for this boundary condition is provided in Appendix Figures A1 and A2.

Step 11B (grouped stability): $\text{eco_choice} \sim \text{frame} * \text{stability_group} + \text{time_pressure} + \text{language} + \text{attitude_z}$

The group-based version (more interpretable visually) shows a similar directional structure:

Frame effect in the baseline (lower stability) group is positive (coef ≈ 1.04 ; $p \approx 0.110$).

The interaction for High stability is negative (coef ≈ -0.88 ; $p \approx 0.240$).

So the extension is best stated as:

The data are consistent with framing sensitivity being more concentrated among less stable decision-makers, but estimates remain noisy under clustered inference. This pattern helps explain why the average framing effect looks unstable in a realistic online setting.

These patterns should be interpreted cautiously and require replication—particularly with larger samples and more balanced stability distributions. The contribution here is therefore framed as exploratory boundary-condition evidence rather than a claim of a new general mechanism or a replacement for prospect-theoretic framing.

Table 5. Decision stability (entropy) as a boundary condition of framing

Variable	Model: framing $\times$ entropy
Intercept	- 0.381
Loss frame	0.611
Time pressure (Yes)	0.375
Language (ZH- S)	- 0.149
Entropy (stability)	0.016
Loss $\times$ Entropy	- 0.412
Environmental attitude	1.109***
Observations	2424
Clusters (participants)	101

Notes: Logit coefficients are reported with clustered standard errors at the participant level. Environmental attitude is standardized. The interaction term tests whether framing sensitivity varies across levels of behavioral stability. *** $p < 0.001$.

(These results should be presented as an “Extension” subsection, after H1–H4 and RT, with the stability plots placed either at the end of Results or in Appendix, depending on how central you want the contribution to be.)

Given the relatively small number of participants and the skewed distribution of behavioral stability in this sample, these results should be interpreted as preliminary. The stability boundary-condition pattern requires replication in larger and more balanced samples to establish its generality. The current findings therefore provide suggestive evidence rather than definitive conclusions regarding the role of stability in moderating framing sensitivity.

This boundary-condition interpretation is consistent with recent work emphasizing heterogeneity and behavioral thresholds in realistic decision environments.

6.7 Robustness and non-randomness checks (Step 12A / 12B)

To avoid overfitting and post-hoc storytelling, robustness checks were pre-specified and intentionally minimal. Overall, the robustness exercises preserve the qualitative structure of the main conclusions: strong attitude–choice linkage, weak/unstable average framing under clustered SE, limited evidence for pressure effects in this implementation, and a stability-related heterogeneity pattern that remains exploratory.

Step 12A — Minimal robustness (three checks)

Extreme RT trimming

Re-estimating the Step 8 choice model after trimming extreme RT observations yields the same qualitative structure: attitude remains strongly predictive ($OR \approx 3.10$, $p < 0.001$), while the key interaction terms remain imprecise. This suggests results are not driven by a small number of extreme RT trials.

Attitude split (directional consistency)

When splitting the sample by attitude, the direction of several effects remains broadly consistent, but precision is limited within subgroups (wide CIs). Notably, in the low-attitude group, language differences remain pronounced (ZH-S substantially lower eco-choice). In the high-attitude group, the pressure main effect trends positive but does not reach conventional significance.

Attention/recall as robustness lever

Given multilingual comprehension constraints, attention failure is not treated as a hard exclusion in the main analysis. Where used as a filter, patterns do not qualitatively overturn the core conclusions, but subgroup sizes shrink and estimates become less stable—consistent with the trade-off between “purity” and external validity in online multilingual samples. Language is therefore treated as a processing/measurement channel and attention checks are interpreted cautiously in this setting.

Step 12B — Non-accidental pattern checks (R1–R3)

To assess whether the stability boundary-condition pattern is a fragile artifact, three checks were run:

Alternative stability proxies (e.g., switch rate, consistency): interaction patterns with frame appear directionally consistent with the gating intuition in at least one proxy specification, though sensitivity to specification is expected given sample size.

Subsample repetition (e.g., excluding extreme RT; restricting to higher valid-trial counts): the frame × stability pattern remains directionally similar and is not driven solely by RT extremes or low-trial participants, but precision remains limited.

Correlation diagnostics: stability is not trivially reducible to the attention flag; however, attention measures in multilingual contexts plausibly mix comprehension and engagement, so they are treated cautiously and used primarily for robustness rather than hard exclusion.

(Place full robustness tables in the Appendix; in the main text, keep this subsection short and disciplined.)

6.8 Results summary

Overall, the evidence points to a restrained but coherent interpretation. The average framing signal is directionally positive but unstable once clustered inference is applied. This pattern is more consistent with conditional relevance than with a universal behavioral effect.

Time pressure, as implemented through instruction, does not produce reliable amplification. This is aligned with the weak behavioral signature observed in reaction time. In this online setting, urgency appears to be heterogeneous and partly engagement-driven, which limits the strength of aggregate effects.

Environmental attitude is the only stable predictor in this dataset. It consistently anchors the baseline direction of sustainable choice, although its moderating role remains imprecise in this sample. Rather than acting as a simple buffer, attitude appears to shape behavioral stability and baseline orientation.

Reaction-time evidence indicates that loss framing increases processing demands even when choice effects remain modest. At the same time, the conflict proxy used here does not function as a monotonic indicator of processing intensity. This suggests that processing

adjustments operate through multiple pathways, including both deeper evaluation and rapid heuristic alignment.

The exploratory extension provides a plausible explanation for the instability of average framing. Framing sensitivity appears more concentrated among individuals with lower behavioral stability. High-stability individuals show weaker responses to contextual variation. This boundary-condition pattern is suggestive rather than definitive, but it offers a structured account of heterogeneity in realistic and multilingual decision environments.

7. Discussion

7.1 Summary of Main Findings

The results indicate that loss framing influences sustainable choice in direction but not with stable precision. This pattern is more consistent with conditional relevance than with a universal behavioral effect. In realistic online environments, contextual signals appear unevenly activated rather than uniformly binding.

Time pressure does not produce a consistent effect. The manipulation likely induced perceived urgency in some participants but did not generate a uniform constraint. This aligns with the reaction-time evidence and suggests that engagement and credibility matter more than formal design.

Environmental attitude shows the strongest and most stable relationship with behavior. Individuals with stronger orientation toward sustainability display more consistent decisions. However, the evidence that attitude directly weakens framing sensitivity remains limited.

At the process level, loss framing is associated with slower responses, indicating increased cognitive engagement. The conflict measure used here does not behave as a monotonic processing proxy. This suggests that decision tension may operate through multiple pathways, including rapid alignment and deeper evaluation.

A central insight concerns stability. Individuals with less stable decision patterns appear more responsive to contextual cues, whereas stable individuals rely more on internal orientation. This helps explain why average framing effects appear weak in realistic environments.

7.2 Framing as a Conditional Mechanism

In much of the traditional framing literature, the effect is implicitly treated as stable. Once a framing manipulation is introduced, systematic behavioral differences are expected to follow. The present results point to a more conditional structure. Framing remains observable, but its behavioral relevance varies across contexts and individuals.

A more plausible interpretation is that framing operates through activation rather than dominance. Its influence depends on whether cognitive resources and engagement allow contextual signals to enter the decision process. Under strong and credible constraints, simplified representations may become behaviorally binding.

This perspective helps explain why empirical evidence in sustainable consumption is often mixed. Environmental decisions are typically embedded in complexity, distraction, and competing goals. Variability therefore reflects processing dynamics rather than theoretical inconsistency. Framing is not absent; it is unevenly activated.

Importantly, this interpretation does not contradict prospect theory. Prospect theory provides the mechanism of reference dependence and loss aversion but leaves open the conditions under which these mechanisms become behaviorally relevant. The present findings suggest that cognitive constraints and decision stability may function as activation thresholds.

From a practical standpoint, framing should not be treated as a universal policy tool. Its effectiveness depends on attention, time structure, and behavioral engagement. In digital environments, where decision speed and cognitive load vary widely, framing interventions are likely to produce heterogeneous outcomes. This shifts the focus from message design alone toward the broader structure of the decision environment.

7.3 Cognitive Constraints and Realistic Decision Environments

A central motivation of this study is to move beyond the assumption that decision makers operate in stable and fully controlled environments. Much of the existing framing literature relies on laboratory settings in which information is limited, attention is relatively focused, and time is sufficient for deliberate evaluation. Under these conditions, framing effects often appear stable.

Real consumption environments differ in fundamental ways. Individuals are exposed to continuous information streams, competing goals, and fragmented attention. Time pressure is not an artificial manipulation but a structural feature of many everyday decisions, including online shopping, limited-time promotions, and rapid product comparison. In such contexts, individuals rarely engage in systematic evaluation across all options. Instead, decisions are frequently made under bounded cognitive resources.

From this perspective, framing cannot be treated as a context-free effect. Its influence depends on the availability of cognitive resources and the structure of the decision environment. The present study therefore introduces time pressure as a key dimension of realism. Rather than isolating framing in simplified settings, the experimental design approximates the speed and variability observed in real consumption.

This pattern is most relevant for external validity. Real-world decision environments rarely impose strict and uniform deadlines. Instead, individuals face fluctuating and partially

self-imposed time constraints. The present results therefore suggest that the effectiveness of time-based behavioral interventions depends critically on the design and credibility of the constraint. This perspective helps explain why amplification effects are often observed in laboratory settings but appear weaker or unstable in field and online contexts. Rather than contradicting resource-based theories, the results support a conditional interpretation in which cognitive constraints interact with attention, motivation, and decision stability.

7.4 Decision Process and Reaction Time

Most framing research focuses on final choices. However, outcome measures alone do not capture the dynamics of decision formation. This study extends the analysis by incorporating reaction time as a behavioral indicator of processing.

Reaction time reflects the structure and intensity of cognitive processing rather than simple response speed. Longer decision times are often associated with increased conflict or deeper evaluation, whereas rapid responses may indicate heuristic or automatic processing. Within a dual-process framework, these patterns can be interpreted as shifts between intuitive and deliberative modes.

The empirical results show that loss framing is associated with slower responses, even when its impact on choice is weak or statistically unstable. This indicates that framing continues to influence internal processing. At the same time, the conflict indicator used in this study does not exhibit a monotonic relationship with reaction time. In some regions of the attitude–framing space, rapid responses appear to reflect strong heuristic alignment rather than low engagement.

These findings suggest that cognitive conflict may not operate as a single measurable construct.

Instead, framing appears to modify processing regimes. The joint analysis of choice and reaction time therefore provides stronger evidence that framing affects decision mechanisms rather than producing purely statistical variation.

7.5 Heterogeneity: Environmental Attitude and Stability

The results further highlight the importance of individual heterogeneity. Environmental attitude functions as a preference anchor that shapes the baseline direction of decision making. Individuals with stronger attitudes display more consistent behavior and weaker

sensitivity to contextual cues. Those with weaker attitudes rely more heavily on situational information, including framing and time pressure.

Beyond attitude, behavioral stability emerges as a central dimension. Stability, measured through entropy across repeated choices, captures systematic variation rather than random noise. This measure appears to reflect a broader processing threshold, including attention allocation, engagement, and sensitivity to external signals.

Framing effects are concentrated among individuals with lower stability. In contrast, high-stability individuals show greater resistance to contextual manipulation. This pattern suggests that stability operates as a gating mechanism that determines whether contextual information influences decision behavior.

This interpretation shifts the focus from average treatment effects toward conditional mechanisms. The effectiveness of framing depends on internal processing regimes rather than uniform bias.

7.6 Integration of Preference and Behavioral Mechanisms

The findings support an integrated view in which stable preferences and contextual influences operate at different layers of the decision process. Traditional economic models emphasize long-run orientation, while behavioral approaches highlight short-run variability. The present results suggest that these perspectives are not competing but sequential.

Preferences continue to shape the baseline direction of behavior. Environmental attitude strongly predicts sustainable choice and reflects underlying value orientation. However, the translation of these values into repeated behavior depends on cognitive conditions and task structure. In realistic and online environments, bounded attention and fluctuating engagement become central.

Contextual signals, including framing and urgency, therefore operate primarily at the level of behavioral expression. They shape how preferences are implemented rather than replacing them. This layered structure helps explain why behavioral variability is common even when value orientation is stable.

In repeated-choice environments, stability emerges as an additional dimension. It captures the consistency with which individuals translate internal orientation into observable action. Lower stability is associated with stronger responsiveness to contextual variation, whereas

higher stability is linked to resistance to external cues. This perspective reframes variability as a structured behavioral feature rather than random noise.

7.7 Policy and Practical Implications

Several implications follow. First, framing interventions should not be treated as universally effective. Their impact depends on cognitive constraints and decision environments. Policies based on message framing may be more effective in contexts characterized by limited attention and rapid decision making.

Second, time structure should be incorporated into policy design. Communication strategies, default options, and digital interfaces interact with time pressure in shaping behavior. Ignoring these interactions may lead to inconsistent outcomes.

Third, heterogeneity across individuals implies that uniform policy approaches are unlikely to be optimal. Different groups may require different strategies depending on preference strength and behavioral stability.

7.8 Limitations and Future Research

Several limitations should be acknowledged. First, the study relies on an online sample. Although such environments are increasingly relevant, replication in field settings would strengthen generalizability.

Second, time pressure is implemented through instruction rather than dynamic countdown. Stronger manipulations may generate clearer effects.

Third, the experimental scenarios focus on a limited range of sustainable products. Future research should examine broader product categories and price structures.

Finally, the current design captures short-term responses. Longitudinal research is needed to investigate how cognitive constraints and stability interact over time and whether behavioral change persists.

8. Conclusion

8.1 Research Question and Contribution

The starting point of this study is a simple observation. Environmental concern has increased, yet sustainable consumption remains limited. The key question is therefore not whether individuals value environmental outcomes, but why these values often fail to translate into action.

The analysis approaches this gap by integrating framing with cognitive constraints. Rather than treating framing as a fixed bias, the study views it as a mechanism whose influence depends on attention, engagement, and decision structure.

8.2 Main Findings

Several consistent patterns emerge from the analysis. Loss framing shows a directional influence on sustainable choice, but the effect remains modest and statistically imprecise across specifications. This is more consistent with conditional activation than with a strong universal bias.

Time pressure, implemented through instruction, does not generate a uniform behavioral constraint. The manipulation appears to affect some participants while leaving others largely unaffected.

Environmental attitude plays a clear and stable role in shaping sustainable behavior. However, its moderating effect on framing is weak in this sample once clustered inference and interaction structures are introduced. This suggests that preference strength and contextual sensitivity operate through partially independent channels.

Reaction-time evidence provides additional insight. Loss framing is associated with slower responses, indicating increased processing demands. At the same time, the operational conflict index does not produce a monotonic relationship with processing time, suggesting that different strategies may dominate across individuals and situations.

Decision stability, measured through entropy, offers a structured explanation for the instability of average effects. Framing sensitivity appears concentrated among low-stability individuals, while high-stability individuals exhibit more consistent behavior. This boundary-condition interpretation remains exploratory but aligns with the broader conditional perspective of the study.

8.3 Theoretical Contribution

These findings support a conditional interpretation of framing. Rather than assuming a universal behavioral bias, framing operates as a context-dependent mechanism shaped by cognitive constraints and engagement. The results shift attention from average treatment effects toward activation and threshold-based perspectives.

Stable preferences remain central in determining long-run orientation. However, their behavioral expression depends on situational constraints and processing regimes. In realistic and noisy decision environments, variability reflects structured interaction rather than random deviation.

Preferences define direction, while contextual signals shape short-run variation. Under this view, behavioral instability is not evidence against rationality but a consequence of bounded processing and heterogeneous engagement.

8.4 Methodological Contribution

Methodologically, the study extends standard outcome-based analysis by incorporating reaction time as a behavioral indicator of processing intensity. This allows the identification of mechanisms that remain invisible in choice data alone.

The use of trial-level data further captures variability and stability across repeated decisions. Rather than treating behavioral fluctuation as noise, the framework interprets variability as informative about decision dynamics. The entropy-based stability index provides a structured way to quantify this dimension.

8.5 Practical and Policy Implications

The results suggest that framing interventions should be context-sensitive. Communication strategies are more likely to be effective in environments characterized by limited attention and rapid decision cycles. Time structure, interface design, and information flow therefore become central policy levers.

Heterogeneity further implies that uniform interventions may produce limited effects. Individuals differ in preference strength and behavioral stability, which affects responsiveness to framing. Targeted and adaptive communication strategies may therefore outperform static approaches in complex consumption environments.

8.6 Future Research

Several directions remain open. First, future work should examine long-term dynamics, including persistence and decay of framing effects. Second, field experiments in natural decision environments are needed to strengthen external validity. Third, adaptive framing strategies that respond to cognitive states and contextual constraints represent a promising research frontier.

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Appendix A. Supplementary figures

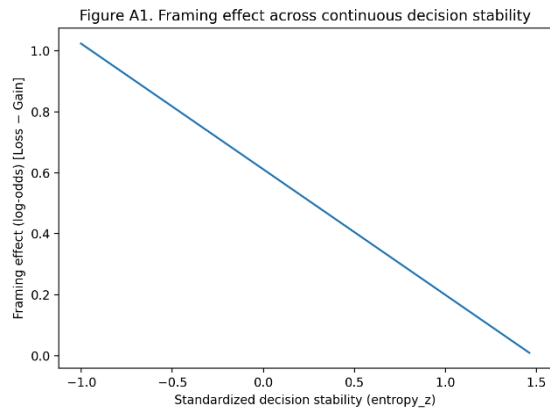


Figure A1. Framing effect across continuous decision stability.

The figure plots the estimated framing effect (Loss – Gain, log-odds) as a function of standardized decision stability (entropy). The downward pattern suggests that framing effects concentrate among less stable decision-makers, consistent with a conditional mechanism interpretation.

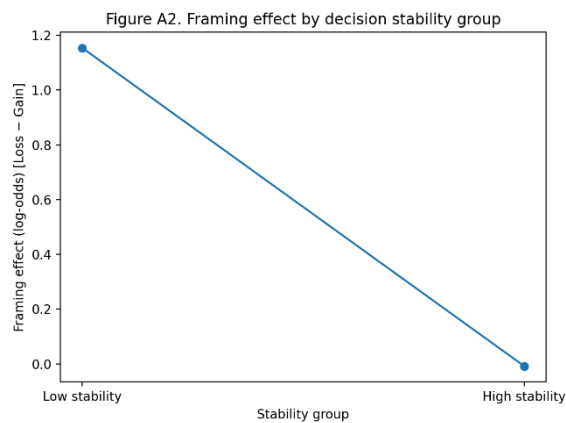


Figure A2. Framing effect by stability group.

The figure compares framing effects across low- and high-stability participants. The stronger effect in the low-stability group supports the interpretation that decision stability moderates framing sensitivity.

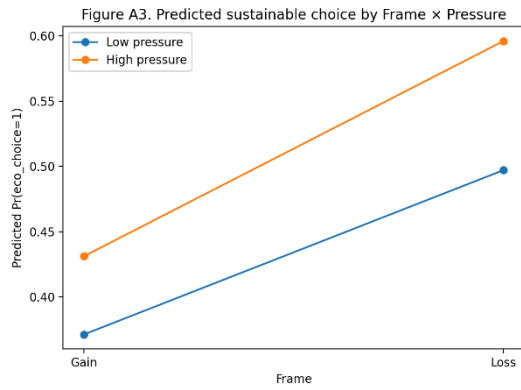


Figure A3. Distribution of entropy-based decision stability.

The distribution indicates substantial heterogeneity in decision stability across participants, supporting the use of entropy as a behavioral heterogeneity index.

Note: Figure A4 omitted for brevity.

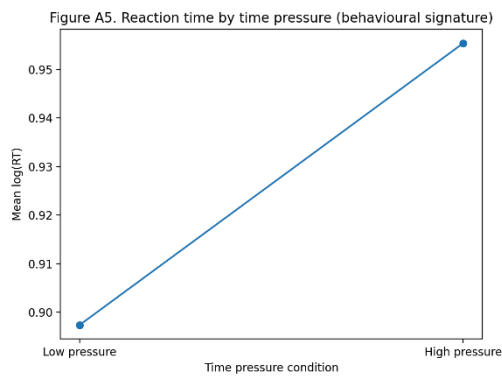


Figure A5. Reaction time by time pressure condition.

The modest difference suggests heterogeneous perception of urgency across individuals, consistent with the weak moderation observed in the regression models.

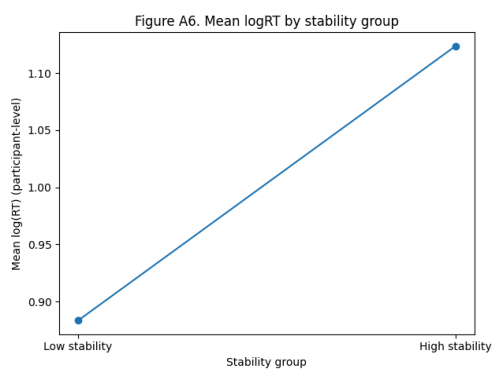


Figure A6. Mean processing time (log RT) by decision stability group.

The figure reports mean log-transformed reaction time across low- and high-stability participants. Patterns are descriptive and interpreted cautiously due to sample size.

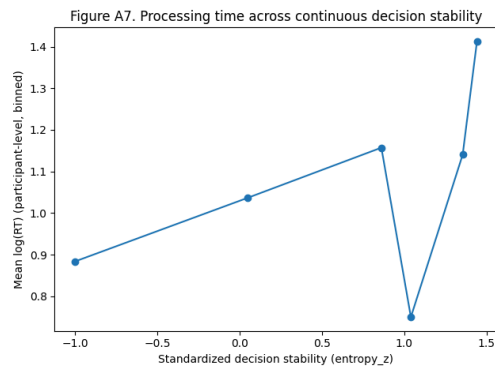


Figure A7. Processing time across continuous decision stability.

The figure plots average log reaction time across bins of standardized decision stability. Results are exploratory and intended to illustrate potential heterogeneity in processing dynamics.